# Validity Is Not Difficulty: Hazard-Preserving Physical 3D World Compilation for Autonomous Driving

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## Abstract

*Driving-world reconstruction systems often conflate two causes of difficulty: a scene may be hard because it contains a legitimate hazardous actor, or invalid because reconstructed geometry contradicts sensor evidence. Treating both as low confidence can obtain superficially safer worlds by deleting precisely the actors that matter. We introduce HARP-3D, a hazard-preserving physical 3D world compiler that separates appearance from collision geometry and artifact validity from behavioral hazard. The compiler aligns multi-frame Actor evidence in canonical coordinates and exposes four surface actions—*KEEP*,* PROJECT*,* COMPLETE*, and* UNKNOWN*—without changing Actor identity, trajectory, or extent. On frozen Argoverse 2, ray-certified compilation eliminates paired free-space violations, reduces target LiDAR depth error from 0.207 to 0.154 m and symmetric Chamfer from 0.254 to 0.175 m, and retains all Actor and hazard state. A target-nearest proximity diagnostic raises hit recall from 49.7% to 69.8%, but only 64.0% of Actors add no visible violation, exposing an aggregate-to-Actor boundary. A nuScenes-only factorized repair-or-abstain head then covers 75.6% of AV2 Actors zero-shot, reduces false repair from 16.6% under always-repair to 9.0%, and obtains 0.182 m selective Chamfer with 0.981 hazard AUROC and zero cross-input shift. On exact-match source Actors it rejects every geometrically harmful repair, yet its selected Actors have 2.28× larger long-tail motion error, showing why repair and trajectory authority must remain separate. Literal first-return audits expose 8.742% source exposure and independently confirm 14.817% on fresh AV2—8.29× its target-nearest proxy—while proving a sharp boundary: deleting points cannot create earlier returns, yet every frozen deletion policy worsens Chamfer. We therefore define consistency in the training target. Actor-canonical child surfaces supervised by GT set, local-plane, first-return, and free-space losses reduce hazardous early returns by 5.123%, improve Chamfer by -6.207 mm, and raise hit recall by +2.759 points on an exposed nuScenes development fold. A ray-normalized categorical head then makes all candidate depths compete for the one-hot GT LiDAR return; relative to those child surfaces it further reduces hazardous early returns by 7.626% and raises hazardous hit recall by +6.603 points, without post-hoc filtering. Its frozen 20-log AV2 evaluation is pending, so we make no zero-shot claim for this learned model. In a retained-source fixed lattice, separate task authority covers 49.4% of action sets and reduces conditional cost from 0.150 to 0.028 without changing Actor or hazard state. We report empirical transfer while explicitly withholding a domain-invariant, conformal, closed-loop, or road-safety guarantee.*

## 1. Introduction

Reconstructed driving worlds are increasingly used for perception analysis, counterfactual replay, and planning evaluation. Their geometry, however, is rarely a literal collision model. A rendered Gaussian may explain image appearance while occupying free space, duplicating an Actor shell, or flickering across time. Conversely, a perfectly valid Actor may be difficult because it cuts in, brakes hard, or crosses the Ego path. A useful world compiler must distinguish these cases: *world validity is not world difficulty*.

This distinction matters because a generic confidence filter has an easy but scientifically undesirable failure mode. It can improve aggregate collision proxies by suppressing uncertain or nearby Actors. Such a world looks safer because legitimate hazards disappeared, not because its physical state became more faithful. We instead ask whether evidence-inconsistent geometry can be repaired while identity, lifecycle, trajectory, size, time-to-collision, and hazard events remain fixed.

We propose HARP-3D, a physical compilation layer between a reconstructed scene and downstream task queries. It represents a world as

$$\mathcal{W}_t = (\mathcal{G}_t^{\mathrm{app}}, \mathcal{S}_t^{\mathrm{phys}}, \mathcal{A}_t, \mathcal{P}_t, \mathbf{q}_t^{\mathrm{val}}), \quad (1)$$

where appearance Gaussians and physical collision surfaces have separate semantics. The compiler uses multi-frame Actor alignment, ray free/occupied evidence, temporal support, and source provenance to select one of four local actions. Unsupported regions become UNKNOWN, never free by default, while Actors remain in the scene.

Our second component factorizes validity from hazard with disjoint inputs and paired counterfactual diagnostics. Changing a clean Actor from ordinary driving to a legal near-miss should not change its repair score; injecting a duplicate shell or temporal pop should not change its hazard score. A selective interface repairs an Actor only when nuScenes-calibrated runtime evidence supports it and otherwise returns the clean query. Our third component links physical state quality to downstream reliability. An Actor residual distribution is projected onto a candidate trajectory boundary, yielding a continuous cost whose conditional density supports arbitrary budgets and multiple horizons.

We additionally test whether the discrete surface compiler can be replaced by a supervision-native Actor-local completion model. A train-only oracle first shows that moving completion geometry can jointly reduce literal early returns and Chamfer. Deletion-style displacement, an implicit field, and a non-learned TSDF then expose opposite failure endpoints. Starting from the strongest deployment-consistent relocation, we define geometry from held-out Actor-canonical LiDAR targets: each completion seed produces a small set of Gaussian-compatible child centers trained by bidirectional surface and literal-ray losses. On an exposed development fold this changes hazardous early returns by 10.220% and Chamfer by -2.902 mm without deleting any child or Actor; external results remain frozen and pending.

Our contributions are:

- a hazard-preserving physical compiler with explicit KEEP/PROJECT/COMPLETE/UNKNOWN semantics, separate appearance/collision state, and a literal first-return audit with an exact deletion-monotonicity boundary;
- a nuScenes-only repair-or-abstain factorization protocol that measures cross-input leakage and reports its AV2 risk–coverage boundary;
- a continuous task-cost density and monotone multi-horizon reliability interface, with an exact-identity audit that keeps physical-repair and trajectory authority distinct.
- a supervision-native, set-to-set Actor surface model that turns sparse completion seeds into GT-supervised child centers while keeping first-return, free-space, and surface losses active through deployment-consistent training.

## 2. Related Work

**Driving-world reconstruction.** nuScenes [4] and Argoverse 2 [34] provide calibrated multi-sensor trajectories. LiDAR is 2.5D ray evidence: points discard free-space intervals [11]. Scene graphs model Actors [23], while 3D Gaussian Splatting models appearance [17]. UniSim targets closed-loop neural sensor rendering [38]; LiDAR4D and DyNFL target space–time LiDAR synthesis and resimulation [35, 42]. These optimize observations, not collision semantics. We ask whether an edited Actor owns a queryable, first-return-consistent surface with explicit unknown space; Gaussian opacity is not occupancy, and our surface preserves provenance.

Neural LiDAR Fields models first returns through a physically motivated ray-weight distribution [12], while categorical depth supervision encourages a sharp metric depth mode [24]. QueryOcc instead samples positive and negative LiDAR queries directly in continuous space [20]. We combine their relevant boundary conditions: native GT rays define supervision, but an Actor-local 3D score is normalized only when answering a ray. This separates learned geometry evidence from the sensor termination distribution and keeps image, motion, and hazard variables outside the geometry head. Ray potentials constrain visibility [27], while evidential occupancy renders its first occupied hit back to LiDAR depth [15, 16]. We separate that literal hit from a target-nearest proxy and name the operator used by every audit. ALSO derives occupied and free supervision directly from the LiDAR sensor origin [3], while SCPNet shows that multi-frame completion labels can contain moving-object traces [36]. DualAD instead carries static and dynamic state in separate streams with explicit ego/object-motion compensation [6]. We use these observations to audit the producer of Actor-local targets and to supervise a continuous physical field; they do not justify deleting an inconsistent point after inference.

Recent 4D formulations make motion a first-class factor rather than a geometry residual. DynamicVGGT equips Gaussians with velocity under scene-flow supervision [10]; DeGO separates rigid offsets from non-rigid deformation [8]; and SelfOccFlow predicts separate static/dynamic signed-distance fields before flow-warped temporal aggregation [32]. These works motivate our separation boundary, not a visual-teacher shortcut: canonical surface, visibility, and temporal deformation must receive distinct physical supervision.

**Learned geometry completion.** SelfOcc treats the reconstructed volume as a continuous signed-distance field and closes training through differentiable rendering [13]. Object-centric occupancy completion instead canonicalizes long LiDAR tracklets and queries an implicit decoder at Actor-relative coordinates [41]. SparseOcc++ separates geometric completion from semantics by regressing an anisotropic scene-completion field on sparse anchors [31]. Gau-Occ initializes semantic Gaussians from a diffusion-completed LiDAR cloud, but its final multimodal head may

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Actor alignment ray FREE/OCC → Physical compiler four local actions → Validity–hazard factorization → Analytic query Actor residuals → Cost density task reliability

Figure 1. Validity is not difficulty. Top: a metadata-locked $2 \times 2$ teaser built from the first non-hazardous (q01-a0) and hazardous (q00-a0) frozen main cases. Within each Actor pair, only the paired synthetic artifact changes; identity, trajectory, extent, camera, and hazard label remain fixed. Bottom: HARP-3D compiles sensor evidence into an Actor-preserving physical world before granting separate task authority. These are calibrated evidence overlays, not photorealistic reconstructions.

still update center, scale, and rotation under occupancy supervision [21]. PoinTr formulates completion as set-to-set translation, while SnowflakeNet grows child points from parent seeds [37, 39]. We combine only these target/set principles: dynamic sweeps are aligned per Actor, observed anchors are immutable, image features cannot update geometry, and ray/surface losses remain active. Our target is not occupancy mIoU: it is the joint reduction of literal early returns without degrading held-out Actor surface distance.

GaussianFormer-2 interprets sparse Gaussians as occupied-region probability distributions [14], while GaussRender adds projective depth and semantic losses alongside native 3D occupancy supervision [5]. Vol3DGS analytically integrates 3D Gaussians to obtain volumetrically consistent transmittance [30]. ShelfOcc likewise argues that metrically consistent native-3D targets are essential in dynamic driving scenes [2]. These results motivate our Gaussian energy, but not unconstrained end-to-end geometry: we keep motion-compensated Actor targets and audit whether ray gradients alter centers or merely inflate support.

Feature 3DGS attaches image-distilled descriptors to Gaussian carriers [44], while Neural Shell Texture Splatting explicitly separates surface geometry from a sampled texture field [40]. We use the same narrow separation principle, not their visual objective: image-trained SH and opacity can be associated with a frozen physical carrier, but cannot update its center or scale.

**Uncertainty and reliability.** Deep ensembles expose useful empirical predictive disagreement [19], and conformalized quantile methods can turn residuals into calibrated marginal intervals under appropriate assumptions [25]. Interval-bound methods can certify a network over a specified input box, including for robust OOD detection [1]; they do not certify that the box models an unseen sensor domain. Driving logs violate many simple exchangeability assumptions through scene correlation and domain shift. We therefore report empirical calibration explicitly, preserve claim boundaries, and use monotone structure only for runtime consistency—not as a formal road-safety guarantee.

Selective prediction couples conditional risk to coverage [9], while learning to defer evaluates the downstream fallback as part of the system [22]. Moreover, abstention can improve aggregate regression yet worsen a predefined subgroup [28]. We therefore report both selected and fallback-composite outcomes and decompose them over the frozen physical hazard stratum; this is descriptive accounting, not subgroup calibration.

**Cross-sensor LiDAR generalization.** LiDAR domain shift includes sensor sampling as well as scene and appearance shift. Source-only subsampling consistency can improve transfer across scan densities [18], while partially accumulated geometry and sequence cues provide a complementary common representation [26]. These results motivate our frozen sparsity intervention, but do not make source consistency a certificate for actor-level repair selection; we therefore retain exact-once source and external evaluations.

**Our distinction.** Prior confidence, occupancy, and reconstruction-quality signals often mix physical invalidity, visibility, and behavioral difficulty. HARP-3D instead defines separate evidence inputs, paired interventions, and failure denominators so that an improvement cannot be obtained by deleting hazardous Actors or relabeling unknown space as free.

# 3. Problem Formulation

Let Actor $i$ have an immutable semantic state

$$A_i = (ID_i, X_{i,0:H}, D_i), \tag{2}$$

containing identity, trajectory, and dimensions. Appearance primitives may be associated with $A_i$, but collision queries consume only an Actor-local physical surface $S_i^{\text{phys}}$ and its known/unknown mask.

We separate two random variables. Artifactness $a_i$ depends on sensor and geometric evidence $E_i$, physical surface state, and provenance:

$$a_i = P(\text{artifact} \mid E_i, S_i^{\text{phys}}, P_i). \tag{3}$$

Hazardness $h_i$ depends on Actor dynamics, map context, and a candidate Ego trajectory $\tau$:

$$h_i = P(\text{hazard} \mid X_{i,0:H}, \tau, \mathcal{M}). \tag{4}$$

The goal is paired conditional invariance, not unconditional statistical independence. For the same clean Actor, a legal safe-to-hazard intervention should preserve $a_i$. For the same behavior, a clean-to-artifact intervention should preserve $h_i$.

For task reliability, let $n_{\tau,t}$ be the local boundary normal, $d_{i,t}(\tau)$ the signed Actor clearance, and $\epsilon_{i,t}$ the Actor residual. We define

$$C(\tau, H) = \max_{i,t \le H} \frac{|n_{\tau,t}^\top \epsilon_{i,t}|}{\max(|d_{i,t}(\tau)|, \varepsilon)}. \tag{5}$$

The runtime interface returns $R(\tau, H, b) = P(C \le b)$ and enforces $\partial R/\partial b \ge 0$ and $\partial R/\partial H \le 0$.

# 4. Method

## 4.1. Actor-Preserving Physical Compilation

For each Actor, we transform LiDAR endpoints, rays, and associated primitives into a canonical Actor frame,

$$x_{i,t}^{\text{actor}} = T_{\text{actor},t}^{-1} T_{\text{ego},t} T_{\text{sensor}} x_t^{\text{sensor}}. \tag{6}$$

The fused representation stores surface support, observed-free intervals, temporal counts, local normals, viewing-direction diversity, and provenance. Compilation is deterministic and local:

- KEEP: stable sensor-supported surface;
- PROJECT: a near-surface primitive contradicts observed free space and is projected to supported geometry;
- COMPLETE: a local hole has sufficient multi-frame and multi-view support;
- UNKNOWN: evidence is missing, contradictory, or insufficient.

Unknown is a first-class state and never silently becomes free. All four actions preserve $(ID_i, X_{i,0:H}, D_i)$. Appearance primitives may attach to the repaired surface using center distance, covariance-normal alignment, depth, and visibility, but physical queries do not read Gaussian opacity.

The physical certificate is visibility-conditioned. For a held-out ray $r = (o, v, d)$ and surface $S$, the target-facing state $\zeta(r, S)$ is one of early, hit, late, or unmatched. In the reverse direction, an output point's state $\eta(x, r)$ is contradicted, supported, occluded, or off-ray unknown. Only the contradicted state places $x$ in observed free space; occluded and off-ray regions remain non-falsified rather than becoming occupied or free. This bidirectional partition makes the failure object inspectable while explicitly withholding a completeness certificate for unseen surface.

For the literal safety audit, define the first valid positive depth within the frozen beam tube as

$$d_S(r) = \min\{s > 0 : o + sv \in S_\epsilon\}, \qquad \min \varnothing = \infty. \tag{7}$$

If a policy only deletes points, $S' \subseteq S$ implies $d_{S'}(r) \ge d_S(r)$. Hence the early predicate $\mathbf{1}[d_S(r) < d^\star - \tau]$ is monotone non-increasing under deletion, as is new-early relative to a fixed query surface. Hit retention and symmetric Chamfer are not monotone under this order: deletion may expose or remove a hit, and it can move the two directed Chamfer terms in opposite directions. This is a geometric certificate for one predicate, not a collision guarantee.

Rather than collapse unknown completion to occupied or free, we can retain a set-valued first return. Let $S_i^K$ contain only build-endpoint anchors and $S_i^P$ additionally contain every GT-supervised completion proposal. Under the explicit coverage assumption $S_i^K \subseteq S_i^\star \subseteq S_i^P$, deletion monotonicity gives

$$d_{S_i^P}(r) \le d_{S_i^\star}(r) \le d_{S_i^K}(r). \tag{8}$$

The endpoints represent possible and known-supported returns and are not thresholded to a single depth. Crucially, $S_i^\star \subseteq S_i^P$ is an empirical completion assumption, not a consequence of the set order; we therefore report target bracketing and unbounded intervals rather than call Eq. 8 a safety certificate.

## 4.2. Learned Evidential Actor Surface Field

V7.1 replaces binary keep/delete on completed primitives with a supervision-native Actor-local geometry action. Every sweep is transformed by its annotated Actor pose before fusion. Build endpoints and ray-transmission intervals define occupied, free, and unknown evidence; held-out target rays never enter the input and only construct the training target $T_i$. Observed KEEP and matched PROJECT points form an immutable anchor set $A_i$.

A deployment-consistent parent model first relocates each completion seed. Because one seed per hole cannot cover a dense target, the point model attaches 4 learned slots to each parent and predicts bounded Actor-frame center residuals and isotropic support scales. The physical point surface used by both training and evaluation is

$$\begin{aligned} x'_{jk} &= x_j^{\text{parent}} + \Delta x_{jk}(E_i), \quad k = 1, \ldots, 4, \\ S_i^\theta &= A_i \cup \{x'_{jk}\}_{j,k}. \end{aligned} \tag{9}$$

Geometry pretraining applies bidirectional Chamfer to $S_i^\theta$ and $T_i$, target-local point-to-plane distance, and a scale target from the target surface's local eight-neighbor spacing. Physical fine-tuning keeps every geometry term active and adds the differentiable first return plus an explicit free-before-hit penalty,

$$\mathcal{L}_G = \mathcal{L}_{\text{CD}} + .25\mathcal{L}_{\text{p2n}} + .1\mathcal{L}_s, \qquad \mathcal{L}_P = \mathcal{L}_G + \mathcal{L}_{\text{first}} + .5\mathcal{L}_{\text{free}}. \tag{10}$$

Measured negative task gradients are projected only during this second stage. Space behind a held-out first hit is not labeled free; unknown is never a deployment deletion mask. Actor identity, trajectory, extent, and hazard state remain immutable interfaces. Image, semantic, motion, and hazard features are absent from the geometry model; a later appearance bridge may update semantics only, with stop-gradient on physical center and scale.

The final development representation separates point geometry from ray termination. Four Actor-local child centers and tangent scales per completion seed are supervised directly by target points, local point-to-plane structure, literal first return, full free-ray intervals, and frame-balanced target coverage. These losses act during training; no predicted child is deleted at deployment. Given an ordered set of query depths $d_1, \ldots, d_K$ inside the Actor box, a shared 3D decoder reads the nearest child latents and relative canonical coordinates to produce logits $a_k$. We define

$$p_k = \frac{\exp a_k}{\sum_j \exp a_j}, \quad F_k = \sum_{j \le k} p_j, \quad \widehat{d} = \min\{d_k : F_k \ge \tfrac{1}{2}\}, \tag{11}$$

and supervise the native LiDAR return with

$$\mathcal{L}_{\text{return}} = -\log p_{k^\star} + \lambda_d \left| \sum_k p_k d_k - d^\star \right|, qquad k^\star = \arg\min_k |d_k - d^\star| \tag{12}$$

The one-hot target is built from GT before training, the decoder sees build evidence only, and the same categorical distribution is used at deployment. Normalization makes $F_k$ monotone with unit mass and prevents multiple zero crossings; it is a representation constraint rather than a post-hoc surface filter. The present protocol assumes a return inside the Actor box and therefore does not claim no-return modeling.

To test whether the ray objective constrains the completed geometry itself, we also remove the query decoder. For immutable anchors and predicted children $S_i = \{(c_j, s_j)\}$, we define a shared Actor-canonical Gaussian energy

$$e_i(x) = \log \sum_j \exp\left(-\frac{\|x - c_j\|_2^2}{2s_j^2}\right). \tag{13}$$

The ray distribution is the softmax of $e_i(o + d_k v)$ over the same metric bins and receives the same one-hot return loss. Hence its only trainable path is through child center and scale; no visibility head can compensate for incorrect anchors. Crucially, native set, plane, scale, and frame losses remain active. The failure-directed final candidate freezes the M8 geometry before applying this energy, because joint ray fine-tuning increases scale while degrading both literal point returns and the energy's own first returns. Observed anchors use a fixed 0.08 m support. This is a development-selected representation whose external evaluation is frozen but pending.

After the geometry-only authority ablation exposes an input-identifiability failure, we retain producer evidence per observed anchor rather than infer a deletion rule from its output. For anchor $j$, the build input stores construction provenance, the source query ray and frame, projection displacement, canonical hit count, temporal and view support, and continuous $m_j^{\text{build}} = (m_F, m_O, m_U)$. Disjoint held-out LiDAR rays construct $m_j^{\text{GT}}$ with free evidence before the endpoint, occupied evidence near the endpoint, and unknown evidence behind the endpoint or without ray support. Field names and optimization paths separate build inputs from GT targets. Neither provenance nor unknown mass is a hard keep/delete decision.

For the order-aware ablation, we replace normalization of pointwise energy by analytic Gaussian transmittance, following the volumetric construction of Vol3DGS [30]. Integrating the normalized line section of each isotropic Gaussian over ray segment $[d_k, d_{k+1}]$ gives optical thickness $\tau_k$. We then form $T_k = \exp(-\sum_{\ell<k} \tau_\ell)$ and termination weight $w_k = T_k(1 - \exp(-\tau_k))$, as in the ordered alpha-compositing used for occupancy supervision by GaussRender [5]. Because the frozen corpus contains a target return inside the Actor box, this ablation conditions $w_k$ on a box return and reports the residual no-return mass separately; it does not claim no-return modeling.

Rigid motion is a separate authority. For annotated pose $T_{it} = (R_{it}, t_{it})$, the deployed field is

$$e_{it}^{\text{world}}(x) = e_i\big(R_{it}^\top(x - t_{it})\big). \tag{14}$$

Thus trajectory changes where an Actor field is queried but cannot deform its canonical centers or scales. We evaluate this expression by inverse-transforming world queries, avoiding loss of local precision in large world coordinates.

Static Background Gaussians are a separate scene component; neither they nor dynamic SH, opacity, or image features enter the physical query. This deployment factorization preserves the frame-balanced GT supervision used to learn $e_i$ rather than introducing a per-frame post-hoc filter.

We make this ownership boundary explicit rather than infer it from an output audit. Inspired by object-transform factorization in scene graphs and neural simulators [23, 38], let the physical state be $P_i = (\{c_{ij}, s_{ij}\}_j)$, the read-only pose be $T_{it}$, and the sibling visual state $V_i$ contain render geometry, SH, and opacity. Geometry is optimized by the GT set, local-plane, scale, frame-coverage, and first-return losses above. Image optimization instead receives $\mathrm{sg}(P_i)$ and updates only $V_i$, where sg denotes stop-gradient. The deployed physical API is

$$Q_{it}(x; P_i, T_{it}) = \log \sum_j \exp\left[-\frac{\|R_{it}^\top(x - t_{it}) - c_{ij}\|_2^2}{2s_{ij}^2}\right], \tag{15}$$

and has no visual argument.

**Proposition 1 (appearance non-interference).** For fixed $(P_i, T_{it})$ and any visual-only update $\Delta V_i$, $Q_{it}(x; P_i, T_{it}, V_i + \Delta V_i) = Q_{it}(x; P_i, T_{it})$ and $\partial Q_{it}/\partial V_i = 0$. This follows directly from the dependency set of Eq. 15, not from a measured post-hoc correlation. Moreover, for a common rigid world transform $g$, inverse querying gives $Q_{gT}(gx) = Q_T(x)$; rigid motion relocates but cannot deform the canonical field. These are ownership and equivariance guarantees, not guarantees that the GT is complete, learned geometry or trajectory is correct, Background collision is modeled, or the resulting scene is safe.

For the appearance interface, each frozen physical center receives only the SH coefficients and opacity logit of its nearest image-trained StreetGS Gaussian with the same Actor identity. The association is computed in canonical coordinates and has no geometry gradient. Appearance center, covariance, and rotation are never copied, so visual support cannot move, resize, delete, or accept a physical primitive. The resulting sidecar is an attribute carrier rather than an image-conditioned geometry model.

For the final evidential surface measure, the categorical logit is induced directly by supervised Gaussian authority. With occupied mass $o_j$ and kernel $\kappa_{kj}$ at depth sample $d_k$, define

$$q(k, j) = \frac{o_j \kappa_{kj}}{\sum_{a,b} o_b \kappa_{ab}}, \qquad p_k = \sum_j q(k, j). \tag{16}$$

This joint measure is invariant to whether responsibility is marginalized over depth or over primitive family. In particular, for the literal safety boundary $b = d_{\mathrm{gt}} - \epsilon$ and the same piecewise-linear CDF used by the renderer,

$$\widehat{d} < b \iff F(b) > \tfrac{1}{2}, \qquad F(b) = F_{\mathrm{anchor}}(b) + F_{\mathrm{child}}(b). \tag{17}$$

Equation 17 is an exact explanation of the deployed decision, not an abstention rule: $F(b)$ is never thresholded to delete primitives or cases.

### 4.3. Validity–Hazard Factorization

The validity encoder receives ray contradiction, surface residual, temporal support, provenance, visibility, and local geometry. Its label is whether the frozen compiler's target Chamfer is no worse than that of the untouched clean query. The hazard encoder receives only TTC, clearance, closing speed, braking, and crossing geometry. Our factorized model uses two structurally disjoint low-capacity heads; a shared-input two-head network is the learned baseline:

$$\mathcal{L} = \mathrm{BCE}(s_{\mathrm{rep}}, y_{\mathrm{rep}}) + \mathrm{BCE}(s_{\mathrm{haz}}, y_{\mathrm{haz}}). \tag{18}$$

We fit features and weights on nuScenes train only. On a disjoint calibration fold, threshold $q$ is the maximum-coverage value satisfying the population false-repair bound

$$\widehat{R}^+(q) = \frac{1 + \sum_{i=1}^n \mathbf{1}[s_i \geq q,\, y_i = 0]}{n+1} \leq \alpha. \tag{19}$$

Accepted Actors use the compiled surface; abstained Actors retain the clean query. Paired input swaps measure whether either score reads the other task's variables. The finite-sample interpretation requires exchangeability; AV2 uses the frozen $q$ only for empirical zero-shot evaluation.

For a frozen stratum $z$, selector $s_i$, query/compiled Chamfer $(q_i, c_i)$, and introduced visible-failure indicator $v_i$, the complete defer-to-query system admits an exact accounting decomposition:

$$\begin{aligned} G_z &= \mathbb{E}_z[s_i(q_i - c_i)] = C_z \mathbb{E}_z[q_i - c_i \mid s_i = 1], \\ M_z &= \mathbb{E}_z[s_i v_i] = C_z \mathbb{E}_z[v_i \mid s_i = 1]. \end{aligned} \tag{20}$$

where $C_z = \mathbb{E}_z[s_i]$ is repair coverage. Population gain and introduced-failure mass are the stratum-share-weighted sums of $G_z$ and $M_z$. Hence Actor-state preservation, conditional risk, and complete-system utility are distinct claims.

### 4.4. Actor Residuals and Boundary-Cost Density

For Actor $i$ and horizon $t$, a low-capacity head predicts a 2D residual distribution $(\mu_{i,t}, \Sigma_{i,t})$. Analytic projection onto $n_{\tau,t}$ gives mean $n^\top \mu$ and variance $n^\top \Sigma n$. Equation 5 converts these residuals and deterministic clearance into a continuous trajectory cost.

We model $y = \log(1 + C)$ with the frozen P182 conditional Gaussian mixture. Its implemented inputs are the Actor residual state $s_{\text{actor}}$, horizon $H$, and deterministic clearance $d_{\text{clearance}}$,

$$p_\theta(y \mid s_{\text{actor}}, H, d_{\text{clearance}}) = \sum_k \pi_k \mathcal{N}(y; \mu_k, \sigma_k^2), \quad (21)$$

which exposes a closed-form marginal CDF for any budget. A low-rank, input-conditioned dependence model combines the frozen marginals across horizons. This factorization separates learned Actor uncertainty from analytic task geometry.

The analytic boundary also exposes deterministic geometry sensitivity. Let $a_j = |n_j^\top e_j|$ and $m_j = \max(d_j, \epsilon)$. For $C(d) = \max_j a_j/m_j$ and a uniform signed-clearance shift $d' = d + \delta$,

$$|C(d') - C(d)| \leq \max_j \frac{a_j|\delta|}{m_j m_j'} \leq \frac{a_{\max}|\delta|}{\epsilon^2}. \quad (22)$$

This follows from the max operator and clipped denominator being 1-Lipschitz; it is an error-propagation bound, not a distributional safety statement.

### 4.5. Monotone Runtime Surface and SceneIR

The final runtime object is a frozen grid over horizon and cost budget with monotone interpolation. Budget CDF values are non-decreasing; prefix-horizon reliability is non-increasing. Groupwise isotonic maps may calibrate predefined conditions on a disjoint source fold, but do not change these structural constraints.

The two authorities compose without implication. The repair gate first chooses the surface representation,

$$\widetilde{S}_i = g_i S_i^{\text{comp}} + (1 - g_i) S_i^{\text{query}}, \qquad g_i = \mathbf{1}[s_i^{\text{rep}} \geq q]. \quad (23)$$

The downstream task separately grants authority only if its multi-horizon reliability satisfies the requested budget, $a_{i,t}(c) = \mathbf{1}[R_i(t, c \mid \widetilde{S}_i, \tau) \geq \rho]$. In particular, $g_i = 1 \not\Rightarrow a_{i,t}(c) = 1$: evidence can support a physical repair while Actor motion remains uncertain.

SceneIR stores Actor identities and transforms, physical surfaces, known/unknown masks, provenance, and validity fields. The AV2 adapter performs only coordinate, unit, timestamp, category, and sensor-opportunity normalization. It does not fine-tune, recalibrate, or change thresholds using AV2 quality.

## 5. Experiments

**Data and protocol.** We train and select models only on nuScenes. Argoverse 2 Sensor validation logs form a zero-shot external test. Before reading any method output, we sort the 150 validation UUIDs and select every fifth log, yielding 20 quantitative and 10 qualitative logs. No AV2 fine-tuning, post-hoc calibration, threshold selection, or failed-scene deletion is allowed. Final nuScenes and AV2 cohorts are evaluated exactly once after the method freezes.

**Baselines.** For physical compilation we compare naive reconstruction, a legacy validity filter, V6.7 inward-ray repair, and HARP-3D. For factorization we compare a shared encoder, a two-head shared trunk, independent encoders, and paired factorization. Reliability baselines include a scalar confidence, independent horizon classifiers, a continuous marginal density, and the full joint multi-horizon surface.

**Metrics.** Physical metrics include free-space violation, ray termination, point-to-surface error, LiDAR depth consistency, temporal jitter, ghost components, and surface completeness. Hazard preservation reports Actor/ID/lifecycle retention, trajectory and TTC shift, and event-count change. Factorization reports artifact/hazard AUROC and AUPRC, clean-hazard false-artifact rate, paired prediction shift, and cross-probe leakage. Reliability reports log likelihood, integrated Brier score, calibration error, Spearman correlation, monotonic violations, and runtime.

**Continuous geometry oracle and learned displacement.** On train-only Actors, a bounded ray/normal displacement oracle reduces hazardous literal early returns by 5.80% while changing mean Chamfer by -3.778 mm and retaining all Actor states. This establishes a feasible continuous action space without deleting completion candidates. The fixed M0 residual model does not reproduce that Pareto point on the 88-Actor source selection set. Hazardous early returns fall by 19.44%, but Chamfer changes by +48.745 mm and hit recall by -10.82 percentage points. It labels 74.51% of completion candidates unknown, so the apparent ray benefit is principally abstention rather than learned relocation. We therefore rejected M0 under the preregistered Pareto rule and evaluate the conditionally specified implicit field next. The field further reduces hazardous early returns by 28.49%, but changes Chamfer by +120.674 mm and hit recall by -14.37 points. It extracts 0 learned surface points on every selection Actor and therefore degenerates to observed hard anchors. We rejected the field, do not tune extraction after selection, and withhold source-final and AV2 claims.

The non-learned B4 evidential TSDF exposes the opposite endpoint on 121 selection Actors. It adds 103,240 surface points, changes Chamfer by -83.961 mm and hit recall by +4.79 points, but hazardous early-return reduction is -106.53%—i.e., exposure more than doubles. Table 2 therefore shows a two-sided failure boundary rather than a winning learned surface model.

We then correct the training target rather than its deployed output. M5 removes the UNKNOWN action and projects the measured first-return/surface gradient conflict, passing all five development gates. Directly matching one Gaussian center to one nearest target point (M6)

Table 1. nuScenes-only selective repair and AV2 zero-shot transfer. False repair is measured over all Actors; coverage is the accepted fraction. Chamfer is for the repair-or-clean-query output. Leak is the maximum mean prediction shift under a paired swap of the other task's input. Each threshold is reused unchanged; "fresh" denotes independently metadata-frozen exact-once cohorts.

| Dataset | Head | Repair AUROC ↑ | Hazard AUROC ↑ | Coverage ↑ | False repair ↓ | Selective CD ↓ | Max leak ↓ |
|---|---|---|---|---|---|---|---|
| nuScenes test | shared | 64.26 | 91.66 | 11.84 | 3.51 | 0.231 | 28.86 |
| nuScenes test | factorized | **64.91** | **98.11** | 3.95 | **0.44** | 0.246 | **0.00** |
| AV2 zero-shot | shared | **77.97** | 92.89 | 55.21 | **3.15** | 0.190 | 23.46 |
| AV2 zero-shot | factorized | 69.96 | **98.05** | **75.55** | 8.99 | **0.182** | **0.00** |
| nuScenes fresh | factorized | **78.23** | **90.60** | 5.69 | **0.00** | 0.214 | **0.00** |
| nuScenes fresh | sparsity-consistent | 74.73 | **90.60** | **26.83** | 2.44 | **0.197** | **0.00** |
| AV2 fresh | factorized | 65.48 | **98.50** | 77.25 | **11.28** | 0.184 | **0.00** |
| AV2 fresh | sparsity-consistent | **67.62** | **98.50** | **83.94** | 12.43 | **0.178** | **0.00** |

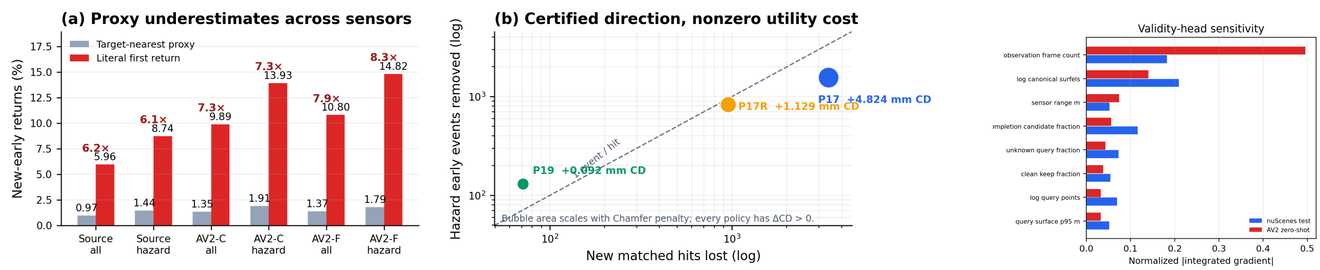


Figure 2. Physical hard evidence and boundaries. Left: target-nearest proximity undercounts literal first-return exposure on source, consumed AV2 (AV2-C), and disjoint exact-once AV2 (AV2-F). Center: deletion certifies non-increasing early returns, but every frozen policy loses hits and worsens Chamfer. Right: attribution localizes the AV2 sensor-opportunity shortcut. These are measurement, structural-direction, and model-sensitivity results—not collision or road-safety certificates.

learns center, tangent-plane, and scale labels but misses the hazardous-ray gate: candidate-bearing Actors have only 17 completion seeds versus 966 target endpoints at the median. M7 instead performs fixed four-child set expansion. Its training Chamfer ratio falls from 0.986 to 0.972 before physical fine-tuning. On the same 66 exposed development Actors, it reduces hazardous/all early returns by 10.220%/8.714%, changes Chamfer by -2.902 mm, and increases hit recall by +2.173 points with complete Actor and hazard retention. M5 gives 5.852% and -1.635 mm on this fold; M6 gives 3.670% and is rejected. These are model-development results because the parent representation previously saw the fold; the frozen AV2 evaluation is reported only after all 20 logs complete.

**Supervision-native geometry and first-return distribution.** We next keep every generated surface and supervise the representation itself. Frame-balanced Actor-canonical seed expansion (M8) reduces hazardous early returns by 5.123%, changes Chamfer by -6.207 mm, and changes hit recall by +2.759 points. Analytic and implicit support variants then expose complementary failures (Table 3): compact support suppresses early returns but misses the surface, whereas an unconstrained direct field restores hits but creates earlier sign changes. Non-negative survival density restores monotonicity but accumulates diffuse pre-hit opacity.

Table 2. Frozen V7.1 continuous-geometry evidence. Early is relative hazardous literal first-return reduction (higher is better); CD and hit are changes from each method's query surface. S1 is a train-only oracle. M5–M7 use the same 66-Actor development fold with prior representation exposure and are not external confirmation.

| Method | Early ↑ | ΔCD (mm) ↓ | Δhit (pp) ↑ | Decision |
|---|---|---|---|---|
| S1 displacement oracle | 5.80% | -3.778 | +2.27 | feasible |
| M0 learned displacement | 19.44% | +48.745 | -10.82 | reject |
| M1 evidential field | 28.49% | +120.674 | -14.37 | reject |
| B4 evidential TSDF | -106.53% | -83.961 | +4.79 | opposite frontier |
| M5 PCGrad relocation | 5.852% | -1.635 | +1.429 | dev pass |
| M6 one-to-one GT center | 3.670% | -1.391 | +1.077 | reject |
| M7 GT set child surface | 10.220% | -2.902 | +2.173 | dev pass |

The categorical model (M18) is the only frozen development candidate satisfying all seven point/ray gates. Relative to M8 it reduces hazardous/all early returns by 7.626%/4.546% and raises hazardous/all hit recall by +6.603/+5.035 points, with 100% observable rays. The clear subset remains negative (early -13.292%, hit -2.678 points), so this is not a uniform dominance or safety result. The fold was exposed to the M8 predecessor; a separately frozen 20-log AV2 run is pending, and no partial external metric is used for model choice.

CVPR
#*****

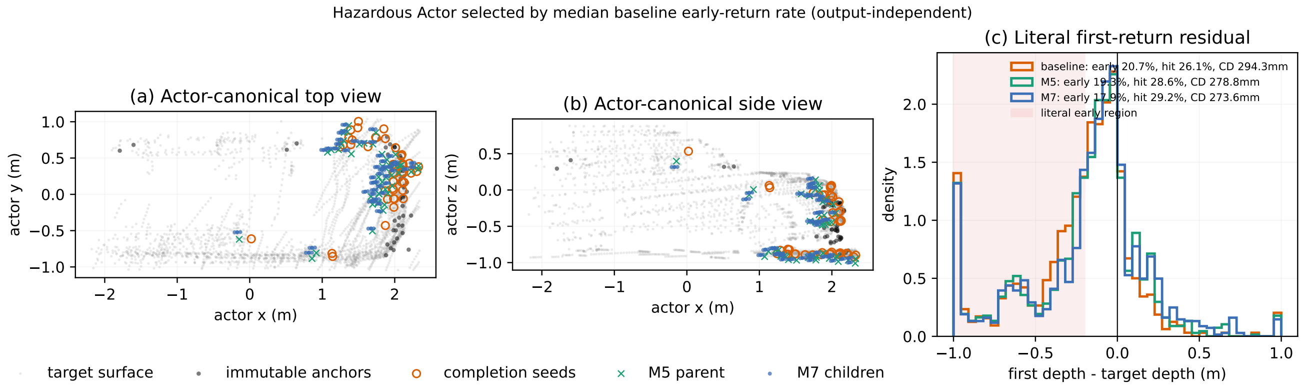


Figure 3. Supervision-native geometry on a hazardous development Actor selected by median *baseline* literal early-return rate, without consulting M7 output. Forty-seven completion seeds expand into 188 child centers while observed anchors remain fixed. For this output-independent case, baseline/M5/M7 early rates are $20.7/19.3/17.9\%$, hit rates are $26.1/28.6/29.2\%$, and Chamfer distances are $294.3/278.8/273.6$ mm. Gray points are held-out GT and never model input.

Table 3. Frozen query-representation ablation on the exposed nuScenes development fold. Early is hazardous relative reduction and hit is hazardous recall change, both versus M8. Only M18 satisfies the pre-registered early and hit gates; no row is an external result.

| ID | Query representation | Early ↑ | Δhit (pp) ↑ | Obs. (%) |
|---|---|---|---|---|
| M13 | blended local planes | -76.68% | -10.57 | 95.90 |
| M14 | compact radial patches | +69.12% | -24.15 | 39.89 |
| M15 | one-sided surface cells | +35.11% | -13.54 | 60.41 |
| M16 | direct signed queries | -11.98% | +2.07 | 93.51 |
| M17 | monotone survival density | +26.52% | -26.14 | 96.18 |
| M18 | categorical first return | **+7.63%** | **+6.60** | **100.0** |

Table 4. Failure-directed coupling diagnostic on the exposed nuScenes development fold. Point early is hazardous relative reduction versus the original completion (negative is worse). Energy columns are absolute hazardous early rate and hit recall. M21 reuses frozen M8 geometry and was selected by this diagnostic; it is not an external result.

| ID | Representation | Point early ↑ | Energy early ↓ | Energy hit ↑ |
|---|---|---|---|---|
| M18 | frozen M8 + learned field | +5.12% | 24.36% | 57.24% |
| M19 | joint geometry + learned field | −4.10% | 24.43% | 58.47% |
| M20 | joint geometry + analytic energy | −3.25% | 19.88% | 57.33% |
| M21 | frozen M8 + analytic energy | +5.12% | 17.42% | 62.07% |

Joint training exposes a stricter identifiability boundary (Table 4). M19 lets the learned field update completed children: Chamfer improves by 2.42 mm versus M8, but the point-surface hazardous early direction reverses from a $5.12\%$ reduction to a $4.10\%$ increase. Its apparently larger field-over-point gain therefore compensates for a degraded reference; absolute field early rate is slightly worse than M18. M20 removes the field decoder, yet predicted scale expands from 0.162 to 0.313 m on average. Its categorical NLL falls while point early returns and the analytic energy both worsen relative to their M8 initialization. We reject both fine-tuning paths.

The diagnostic nevertheless identifies a simpler candidate: frozen M8 geometry followed by the same decoder-free energy gives $17.42\%$ hazardous early rate and $62.07\%$ hit recall, without modifying a Gaussian after its native-3D supervision. This M21 row is selected on the exposed development fold, so it is not an independent result. We freeze its representation and the existing 20-log AV2 cohort before reading any partial quality; zero-shot claims remain pending.

**Zero-shot Actor-local surface evidence.** Following the dynamic-Actor canonicalization boundary used by NeuRAD [33], we transform ego-frame AV2 LiDAR points inside each annotated cuboid into its Actor frame. Unlike appearance-oriented seed initialization, we never mirror points: every collision surfel must retain sensor and temporal support. We fuse two of every three frames at 0.12 m and reserve the remaining frames as unseen target geometry. Across 1,046 Actors and 1.075M stable surfels, fusion reduces the mean target-to-surface distance from 0.864 m to 0.131 m and raises recall at 0.20 m from $26.16\%$ to $85.72\%$ (Table 5).

The paired corruption atlas contains clean, observed-free ghost, duplicate-shell, temporal-flicker, and supported-hole probes without changing Actor identity, trajectory, size, or hazard. It yields $100.0\%$ artifact recall, $0.00\%$ clean-hazard false-artifact rate, and $100.0\%$ Actor/hazard retention. These perfect paired scores verify the deterministic compiler contract and input separation; they are not reported as natural-artifact generalization.

| AV2 Actor surface | Dist. (m) ↓ | Recall (%) ↑ |
|---|---|---|
| Single observed frame | 0.864 | 26.16 |
| Multi-frame canonical surfels | **0.131** | **85.72** |

Table 5. Zero-shot held-out LiDAR support on the frozen 20-log AV2 quantitative cohort (1,046 Actors, including 241 hazardous Actors). The comparison uses real sensor points; paired synthetic corruptions are evaluated separately as a compiler contract.

| Surface | Free (%) ↓ | Recall (%) ↑ | CD (m) ↓ |
|---|---|---|---|
| Single-frame input | 100.0 | 53.36 | 0.254 |
| Nearest canonical | 84.91 | **62.60** | **0.168** |
| Ray-certified | **0.00** | 59.56 | 0.175 |

Table 6. Frozen zero-shot AV2 quantitative result. Nearest-surface projection fits geometry best but violates matched-ray free space; ray-certified projection accepts a small Chamfer cost. Later held-out frames are target-only.

**Four-action physical compilation.** We next freeze one single-frame query per Actor and reserve all later held-out frames as target-only geometry. The compiler reads only the build surface and query evidence. A nearest-canonical projection initially reduces Chamfer to 0.168 m, but ray-specific evaluation reveals 84.91% early terminations in the paired observed-free interval. Our final operator is ray-certified: a primitive with direct LiDAR provenance projects to its matched observed termination; without one it becomes UNKNOWN. Across 433 quantitative Actors (147 hazardous), this reduces paired free-space violations from 100.0% to 0.00%, target depth error from 0.207 m to 0.154 m, and Chamfer from 0.254 m to 0.175 m, while raising ray-termination consistency from 56.94% to 64.38% (Table 6). Actor identity, trajectory, extent, and hazard labels are retained for 100.0% of Actors. The zero free-space rate is a paired contract result; depth, ray termination, and Chamfer use independent target-only LiDAR.

**Frozen camera evidence and an exposed boundary.** We project each pre-registered qualitative Actor back into synchronized AV2 ring-camera frames using the official calibration and motion compensation. Camera identity maximizes only the number of visible query LiDAR returns, with a fixed camera-order tie break; RGB is decoded afterward and every case is retained. All 30 cases have calibrated RGB, sparse depth, four-action, and before/after video evidence; the median/minimum visible query counts are 82/14 (Fig. 1). Importantly, the frozen set exposes a limit: although mean Chamfer improves, 17.19% of all 634 Actors worsen individually. Thus P3 establishes aggregate zero-shot geometry, not a per-Actor repair certificate; this motivates a selective "repair or abstain" validity head trained and calibrated only on nuScenes.

**Visibility-conditioned physical evidence.** We further partition target-only LiDAR evidence with a target-nearest proximity operator into early, matched hit, late, and unmatched; reverse surface evidence is supported, contradicted by observed free space, occluded, or off-ray unknown. Unknown is never counted as free. On all 634 Actors, query-to-compiled hit recall rises from 49.69% to 69.83%, early termination falls from 3.395% to 2.868%, and visible-surface precision rises from 99.61% to 99.77% (F-score 0.663→0.822). Yet only 64.04% of Actors add no visible violation. An independent 20-log read reproduces hit 49.72→68.99%, early 3.448→2.931%, and F-score 0.663→0.815, while only 62.72% add no visible violation and 19.50% worsen in Chamfer. Provenance attribution explains the trade-off: completion supplies 95.12% of new early rays but 14.96 new hits per such ray, while only 13.49% of contradicted output points come from completion. We therefore retain the net 9,439-ray reduction, reject a universal certificate, and route Actor-level authority through abstention rather than delete completion. These frozen values localize errors around each target return; they are not literal first-return rates, which are audited below over every candidate surface point in the beam tube.

**nuScenes-to-AV2 selective factorization.** The strict multi-frame surface contract yields 29/56/228 Actors for nuScenes train/calibration/test; we retain the small training set rather than loosen admissibility. On nuScenes test, factorization is non-inferior to the shared head for repairability (64.91 vs. 64.26 AUROC) and raises hazard AUROC from 91.66 to 98.11, while maximum paired cross-input shift falls from 28.86% to 0.00%. The conservative threshold covers 3.95% with 0.44% population false repairs.

Without AV2 fitting, the same selector covers 75.55% of 634 Actors and reduces false repair from 16.56% under always-repair to 8.99%. Clean-query/always/selective Chamfer is 0.258/0.177/0.182 m. Thus abstention costs only 5 mm mean Chamfer relative to always-repair while nearly halving false repairs. Hazard AUROC is 98.05, hazard coverage is 92.17%, and factorized cross-input shift remains zero. However, coverage changes from 8.93% on nuScenes calibration to 75.55% on AV2; Table 1 therefore supports empirical transfer, not a cross-domain conformal guarantee.

**Interpretable safety envelope.** We retain the model, standardizer, thresholds, and all Actor rows and evaluate 21 fixed coverage levels without refitting or recalibration. The factorized score shifts from mean/median .761/.945 on nuScenes calibration to .978/.999992 on AV2 (Wasserstein .217, KS .704), explaining the operating-point coverage jump. Integrated Gradients [29] relative to the nuScenes-training mean attributes 18.25% of absolute validity sensitivity to observation-frame count on nuScenes test but 49.53% on AV2 (Fig. 2, right). The complete fixed-threshold risk–coverage and geometry–coverage curves remain in the supplement. Structural cross-input derivatives remain exactly zero: task factorization prevents hazard leakage, but does not make validity inputs sensor invariant. We therefore preserve the empirical P4 result while rejecting the stronger interpretation of a domain-invariant selector.

We then propagate fixed 0.10 nuScenes-training-standard-deviation feature boxes through the frozen validity MLP in FP64. Only 33.33% of nominal nuScenes-test selections are robust to the full box, versus 91.23% on consumed AV2. Yet 10.76% of those AV2 robust selections are target false repairs: local threshold invariance can stably preserve a domain-shifted error. Sensor-opportunity perturbations create mean logit widths of 1.550/1.285 on nuScenes/AV2, and query-point count is the largest single-feature interval source for 83.33/95.11% of Actors. This actor-level certificate makes the shortcut inspectable, while explicitly separating network stability from repair correctness.

**Fresh exact-once selector boundary.** Before this read, we exclude every P4 train/calibration/test scene and freeze 20 of the remaining 106 locally available scenes by equally spaced official scene index. The 20 scenes yield 123 Actors and are read once, with no replacement or model/threshold update. Motivated by source-only LiDAR sparsity consistency [18], a pre-frozen candidate increases coverage from 5.69% to 26.83% and improves selective Chamfer from 0.214 to 0.197 m. It nevertheless lowers repair AUROC from 78.23 to 74.73 and fails the pre-registered 2-point non-inferiority margin (Table 1). A descriptive common-coverage audit follows the selective-classification AURC formulation of Zhou *et al*. [43]: candidate/P4 AURC is 0.126/0.106 (lower is better), confirming global ranking loss rather than only threshold mismatch. We retain the P4 selector; the recovery remains a negative ablation.

On 523 exact-once AV2 Actors, the order reverses: candidate/P4 AUROC is 67.62/65.48 and Chamfer 0.178/0.184 m, but false repair is 12.43/11.28%. External gates pass, yet this source/external reversal is cohort-dependent rather than universal; P4 remains primary.

**Physical authority is not a visibility certificate.** We exact-join the same 523 fresh AV2 Actors to the independently frozen bidirectional visibility audit, without changing either score or threshold. P4 selects 77.25%; its Chamfer-worsening rate falls from 19.50% under always-repair to 14.60%. Selected visible-failure risk changes only from 37.28% to 36.39%, and its one-sided 95% Wilson upper bound 40.40% remains above the always-repair point risk. Safe-visible AUROC is 0.533, and abstention captures 24.62% of visible failures. Thus P4 controls its Chamfer target but does not confidence-separate visibility risk. Requiring no COMPLETE output and conjoining with P4 lowers visible failure to 20.16% at 23.71% coverage, but Chamfer worsening rises to 43.55% and hazard coverage collapses to 3.52%; observed provenance is therefore visibility evidence, not jointly valid repair authority.

A dedicated nuScenes-only visibility head confirms this mismatch rather than solving it. Safe-visible AUROC is 0.753 on source test and 0.625 on AV2. Visibility-only selection reaches 11.63% AV2 visible risk at 8.22% coverage; conjunction with P4 reaches 10.26% (95% upper 20.98%) at 7.46% coverage. Yet dual Chamfer-worsening is 35.90% and hazard coverage is 3.52%. We reject the head: low target-specific risk obtained by losing geometry and hazardous Actors is not a safety authority.

The complete defer-to-query system preserves 100.0% of Actors and separates conditional risk from population utility. P4 has 28.11% population introduced-visible failure with 0.083 m composite Chamfer gain; P6-C has 30.78% and 0.089 m, remaining an AV2-only frontier point because its fresh-nuScenes ranking reverses. P4-and-visibility has only 0.76% introduced failure but 0.0012 m gain. In contrast, the provenance policy has 4.78% introduced failure and -0.0009 m gain, so query-only strictly dominates it. Low selected risk alone therefore does not establish fallback-system utility.

Exact stratification exposes where this utility is obtained. Hazardous Actors are 27.15% of the cohort; P4 selected visible risk is 47.37%/31.00% on hazardous/clear Actors versus 46.48%/33.86% under always-repair. Only 3 of its 48 removed failures are hazardous, yet they carry 42.86% of introduced failures and 56.42% of gain. Fresh target-nearest action attribution localizes the mismatch: hazardous/clear proxy new-early rates are 1.912%/0.980% (1.951×); COMPLETE causes 94.70% of new early returns and 99.94% of new hits, whereas KEEP causes 95.68% of surface contradictions. P4's hazardous rate is 1.922% (1.006× always), and P6-C is 1.001×. State preservation therefore does not imply hazard-stratum risk reduction or suppression of its completion-driven mechanism.

**Literal first-return correction and safety–surface boundary.** The preceding frozen action audit uses the target-nearest point as a proximity attribution. Following raw-ray occupancy evaluation [15], we separately re-audit fixed source and consumed AV2 surfaces with Eq. 7. The source total/hazard proxy rates 0.966/1.436% rise to 5.956/8.742%. P17/P17R/P19 reduce hazard exposure to 7.777/8.230/8.662%, but each worsens Chamfer. Thus the

Table 7. Frozen V6.7 reliability evidence inherited without V7 refitting. P183 has 10 fresh logs; source held-out has 3,742 trajectories; P201 has 1,846 trajectories from 10 fresh logs. $\Delta$B/$\Delta$E are Brier/calibration-error reductions. P346 reuses P201 and remains development evidence.

| Object | Split | $\Delta$B $\uparrow$ | $\Delta$E $\uparrow$ | cov./unsafe $\uparrow\downarrow$ |
|---|---|---|---|---|
| density | P183 fresh | 28.48 | 69.38 | – |
| joint | source held-out | 16.97 | 71.85 | – |
| joint | P201 fresh | 17.52 | 53.37 | – |
| P346 | P201 reused | – | – | 32.84/9.09 |
| P346 | source held-out $H$ | – | – | 29.42/26.71 |

Table 8. Frozen physical–trajectory interface on exact-match nuScenes test Actors. Cost and error are row-level 90th percentiles over retained V6.7 outcomes; FS/flip gives event counts. Physical selection is not trajectory authority.

| P4 group | Actors | q90 cost $\downarrow$ | q90 error $\downarrow$ | FS / flip $\downarrow$ |
|---|---|---|---|---|
| selected | 5 | 0.283 | 1.577 | 0 / 0 |
| abstained | 83 | 0.102 | 0.693 | 7 / 47 |
| geometric harm | 23 | 0.076 | 0.543 | 0 / 4 |

certified early direction has nonzero surface cost.

Consumed AV2 first shows hazard exposure 1.912→13.926% (7.29×). A disjoint exact-once 10-log read then independently confirms all/hazard 1.370/1.788→10.801/14.817% (7.89/8.29×). All 70,220 fresh literal new-early rays are COMPLETE; completion supplies 98.398% of 86,883 new hits, only 1.237 hits/early. Figure 2 (left and center) visualizes both findings. They transfer metric failure and provenance, not model, policy, collision, or road-safety validity.

**Frozen continuous and joint reliability.** We inherit the V6.7 reliability objects without V7 refitting. On 10 previously unread nuScenes logs, the five-component log-cost density reduces mean integrated Brier and marginal calibration error by 28.48% and 69.38% relative to the frozen monotone-CDF control. A conditional four-horizon copula improves joint-event Brier by 16.97% on source-held-out trajectories and by 17.52% on an independently frozen 10-log cohort; calibration-error reductions are 71.85% and 53.37% (Table 7). The downstream P346 authority reaches 32.84% q90 coverage at 9.09% unsafe rate on that consumed fresh cohort, but source held-out-horizon risk rises to 26.71%. We therefore treat density, dependence, and task authority as separate inherited evidence, not as a V7 repair effect or a formal cross-domain calibration claim.

**Physical repair is not trajectory authority.** We exact-join P4 identities to retained V6.7 multi-horizon outcome rows through the official nuScenes scene order and DriveStudio instance tokens. Only 5 P4-train Actors from two scenes align, so we reject joint fitting rather than recycle calibration/test Actors. The frozen descriptive audit instead contains 88 test Actors and 39,285 rows (Table 8). P4 selects 5 Actors and abstains on all 23 whose compiled Chamfer is worse than the query; selected-and-harmful is therefore zero. Its 1,781 selected rows contain no false-safe or decision-flip events. Nevertheless, selected q90 motion error is 1.577 m versus 0.693 m for abstained Actors (2.28×), reaching 4.246 m at 3 s. The physical gate protects representation quality but cannot replace the downstream reliability authority. These are retained source outcomes, not a causal effect of repair or an execution of the reliability model.

Table 9. Retained-source $2 \times 2$ composition on 88 exact P5 Actors (32 hazardous), all retained. Coverage/cost/unsafe are over fixed two-action sets (all sets for "none"). Equal B0/B1 and B2/B3 action metrics are the intended physical/task non-interference contract, not a causal planning effect.

| Arm (surface / authority) | CD $\downarrow$ | cov. $\uparrow$ | cost $\downarrow$ | unsafe $\downarrow$ |
|---|---|---|---|---|
| B0 query / none | 0.229 | 100.00 | 0.1497 | 16.45 |
| B1 HARP-3D / none | **0.223** | 100.00 | 0.1497 | 16.45 |
| B2 query / P346 | 0.229 | 49.43 | **0.0279** | **1.15** |
| B3 HARP-3D / P346 | **0.223** | 49.43 | **0.0279** | **1.15** |

We also stress the analytic geometry-to-cost bound on 575,596 retained profiles under 6 fixed $\pm 0.05/0.10/0.20$ m shifts. FP64 evaluation has zero violations. At $-0.20$ m, full-source mean/q99 absolute cost shift is 0.0292/0.1450, while only 0.841% of rows cross signed clearance. P4-selected versus abstained mean shift is 0.00085/0.00456 (5.38× more stable) with no selected crossings. Selected Actors can therefore be motion-difficult yet geometrically stable, further separating the three axes of repairability, geometry sensitivity, and trajectory uncertainty.

**Minimal composed-authority utility.** Task-relevant monitoring should propagate prediction error to the queried planning cost [7], rather than treat all prediction errors alike. We therefore form a retained-source $2 \times 2$ audit over query/HARP-3D surfaces and no-task/frozen-P346 authority. The 88 exact Actors (32 hazardous) yield 1,228 fixed two-action sets at the P346 artifact's held-out 2.5 s horizon. Frozen 90% requested reliability authorizes 49.43% of sets, reducing conditional mean actual cost from 0.1497 to 0.0279 (81.37%) and unsafe-set rate from 16.45% to 1.15%. Independently, selective HARP-3D lowers mean surface Chamfer from 0.229 to 0.223 m with all Actors and hazards retained (Table 9). Physical branches have identical action outcomes by construction: this validates interface composition, not closed-loop benefit from repair.

**Rigid dynamic–static composition audit.** We identity-match 12 scene-0230 Actors (7 hazardous) between the frozen physical model and a read-only StreetGS checkpoint. Eleven are moving, and the retained trajectories contain 1,677 valid frames. Across fixed first/middle/last samples (36 Actor–frames), 5,791 physical Gaussians are transported by the checkpoint SE(3) poses while 106,807 appearance-owned and 1,140,862 Background Gaussians remain outside the physical query. The maximum inverse-query energy residual is $3.40 \times 10^{-14}$ and the maximum pairwise-distance residual is $1.24 \times 10^{-14}$ m, despite 126.28 m maximum translation. A first implementation that formed float32 pairwise distances directly in world coordinates produced a spurious 3.125 cm residual; we retain it as a numerical failure and use the mathematically equivalent inverse-query form, without changing geometry, cohort, or tolerance. This audit verifies deployment composition, not a new surface-accuracy or safety gain.

The geometry-locked appearance bridge assigns same-Actor StreetGS SH/opacity to all 5,791 carriers (mean Actor-level association distance 0.127 m) without copying visual geometry. Its physical surface improves Chamfer by 7.16 mm and hit recall by 3.19 points in this scene, but worsens all/hazard early returns by 1.60/0.60%. We therefore treat it strictly as evidence that image-trained attributes can be attached without changing geometry, not as a favorable appearance or physical example.

We then execute the bridge with the native StreetGS rasterizer on a view frozen solely by 3D frustum visibility (Fig. 4). Replacing the selected Actor's 32,522 visual primitives with 309 physical carriers preserves the renderer interface but reduces footprint PSNR from 28.59 to 17.22 dB ($-11.37$ dB) and full-image PSNR by 3.99 dB. The blur and ghosting reject naive SH/opacity transfer as an appearance solution. We do not tune scale, opacity, primitive count, or view after observing this result, and the visual failure does not alter the GT-supervised physical carrier.

To separate a missing learning signal from insufficient carrier capacity, we next freeze every center, scale, rotation, trajectory, Background parameter, and non-target Actor attribute, and optimize only the selected Actor's SH coefficients and opacity for 320 steps on eight geometrically preselected views. On six disjoint held-out views, footprint PSNR rises from 15.86 to 16.94 dB; all six views improve. The corresponding original StreetGS Actor reaches 25.35 dB, leaving an 8.41 dB gap. Thus image supervision can train attributes without changing physical geometry, but appearance-only optimization does not rescue the 309-primitive one-carrier representation. We do not convert this single-scene capacity probe into an appearance or generalization claim.

We finally derive nine fixed tangent surfels per physical carrier using only an 8-neighbor PCA frame, the supervised carrier radius, and a 2 cm normal thickness. These 2,781 visual-only primitives never enter the physical query and their geometry receives no image gradient. Held-out PSNR reaches 17.57 dB, 0.63 dB above the one-carrier result, but the gap to the original remains 7.78 dB. Moreover, only the largest-footprint held-out view improves over the one-carrier model; the other five regress by 0.10–0.31 dB. A fixed coarse–fine hierarchy then combines the 309 parents and 2,781 surfels, assigning half the optical depth to each level before attribute-only refinement. It reaches 17.73 dB and improves all six views over the one-carrier result (median $+0.47$ dB, range $+0.24$ to $+1.07$ dB). These views were exposed before designing the hierarchy, so this is mechanism evidence rather than a fresh appearance benchmark; the 7.62 dB residual gap remains substantial.

We also test whether average geometry losses hide safety-relevant tails. Starting from M8, M29 adds a GT-only bidirectional loss equal to the mean of the worst 10% nearest-point distances, normalized per Actor, while retaining all set, plane, scale, frame, and literal first-return losses. On the 66 exposed development Actors, target-to-surface tail distance improves by 0.924 mm, frame-mean distance by 0.843 mm, and Chamfer by 0.418 mm relative to M8. However, surface-to-target tail distance worsens by 0.362 mm and hazardous early-return rate rises from 26.37% to 26.69%. The geometry and ray gradients conflict in 86.6–90.6% of batches. This paired asymmetry rejects a single symmetric tail objective as a joint solution: increasing GT coverage can still move a small set of generated surfaces into unsupported free space. We stop without sweeping the tail fraction, weight, or seed.

M30 then preserves this conflict as the return interval in Eq. 8: immutable build anchors define the known-supported endpoint and anchors plus all M8 children define the possible endpoint. Set ordering has zero violations over 99,208 rays. Yet only 56.69% of GT returns are empirically bracketed, 47.03% of upper endpoints are unbounded, and the Actor-mean finite-width q90 is 0.326 m (0.343 m for hazardous Actors). More fundamentally, anchor-only queries already produce 19.58% early returns. The interval is therefore an honest epistemic interface, but not yet a tight collision certificate; the residual failure precedes completion and points to cross-frame canonical supervision or pose/visibility inconsistency.

M31 replays the exact 66 Actors through their original raw or processed producer and labels every anchor first return by construction provenance. It exactly recovers the 19.58% anchor early-return rate. Of 19,424 early rays, 66.70% terminate first on a kept query endpoint and 33.30% on a projected endpoint. Projected points form only 13.36% of anchors, so they are overrepresented among contradictions by a factor of 2.49, but removing them cannot address the kept majority. The existing evidence rule changes raw-query early returns from 18.25% to 17.62%; rejected query points alone account for only 1.29% of rays. Hazardous versus clear rates are 20.02% versus 17.43%, while moving versus quasi-static rates are 19.63% versus 19.26%, and the target-frame trend is non-monotonic. Thus neither completion, projection, UNKNOWN filtering, nor motion alone explains the hard-anchor contradiction. We close post-hoc anchor filtering and instead treat free/occupied/ unknown ray evidence as supervision of physical authority.

We next freeze every M8 center and scale and train only a continuous FREE/OCCUPIED/UNKNOWN mass per Gaussian from primitive ray votes and the same whole-ray categorical target. This removes both M19's decoder compensation and M20's scale shortcut, and occupied mass weights Gaussian energy continuously without thresholding or deletion. The model reduces evidential cross-entropy from 0.963 to 0.913, but changes categorical NLL by only 0.003. Relative to unit authority, all/hazard/clear early returns worsen by 0.336/0.338/0.322 points; hit recall improves by 0.487 points. Predicted mean anchor mass is $(m_F, m_O, m_U) = (0.127, 0.504, 0.369)$, indicating a learned global prior rather than primitive-specific contradiction resolution. We reject this geometry-only authority head without tuning. M31's missing source-frame and multiview evidence must be retained by the corpus producer before authority can be identifiable from build-time inputs.

M33 performs this producer-side repair without changing the existing corpus. Exact replay aligns all 1,004 Actors and 297,535 anchors; 13.75% are projected. Build and held-out occupied masses have correlation 0.514, while direct three-state argmax agreement is only 43.41%, so build evidence is informative but is not itself an authority label. Moreover, 58.72% of anchors receive both free and occupied held-out votes, and projected anchors have higher mean held-out occupied mass than kept anchors (0.466 versus 0.425). These data reject provenance deletion and motivate a supervised continuous mapping from producer evidence to frozen Gaussian authority.

M34 learns that mapping while freezing every center, scale, trajectory, and completion-child authority. The predicted anchor occupied mass correlates 0.462 with held-out mass, establishing that the producer evidence repairs M32's identifiability failure. Nevertheless, weighting the spatial Gaussian energy and normalizing it across ray samples increases all/hazard/clear early returns by 0.223/0.177/0.447 points and reduces hit rate by 0.753 points. This separates two failure levels: the model can predict which anchors receive occupied evidence, but the normalized energy CDF is not an ordered first-hit process. We reject M34 without tuning and replace this composition with prefix transmittance rather than changing the supervision or deleting anchors.

M35 trains the same evidence head through analytic segment integration and prefix transmittance. Learned authority moves the ordered operator in the right direction: relative to unit transmittance, early returns fall from 42.29% to 35.27% and hits rise from 50.94% to 56.71%. However, both remain far behind the original unit-energy baseline of 18.19%/62.28%; hazardous early returns are 18.77 points worse. Unit and learned residual no-return masses are 0.320 and 0.399 despite conditioning evaluation on an Actor-box return. The failure is therefore optical parameterization, not ordering: summing unit line integrals over a dense, overlapping primitive set makes opacity depend on sampling density. We reject the current transmittance field and next separate anchor and completion-child optical contributions before defining a GT-calibrated surface measure.

The frozen M36 decomposition localizes that measure error. Learned anchors alone yield 18.39% early returns, only 0.20 points above the original baseline, whereas unit-opacity completion children alone yield 37.46%. Combining those children with learned anchors raises early returns to 35.27%. Thus a global optical scale or further anchor tuning would target the wrong component: the immediate missing supervision is completion-child FREE/OCCUPIED/UNKNOWN authority from its parent build evidence, kept separate from the observed-anchor head.

M37 supplies exactly that separate supervision while freezing every center, scale, trajectory, and the M35 anchor head. The child occupied mass correlates 0.431 with held-out ray evidence. Relative to unit-opacity children, learned child authority reduces early returns from 35.27% to 23.92% and increases hit recall from 56.71% to 65.59%. This is strong evidence that completion opacity is learnable from producer-side context rather than a post-hoc rule. It is not yet a physical pass: relative to the original unit-energy reference, all/hazard/clear early returns remain higher by 5.73/6.35/2.69 points, even as hit recall improves by 3.30 points. The ordered categorical NLL is almost flat. We therefore reject M37 and identify a loss-level omission: the observed interval before a LiDAR first return must itself supervise survival, rather than being represented only indirectly by competition with the return bin.

M38 adds that native supervision as the negative log survival contributed by completion children before the observed return, with all geometry and the anchor head frozen. It lowers held-out pre-hit child optical thickness by 32.0%, reduces early returns by 2.41 points relative to M37, and simultaneously raises hit recall by 1.33 points. Nevertheless, early returns remain 3.32 points above the original energy reference. The child evidential cross-entropy worsens from 0.934 to 1.058 as mean occupied mass falls from 0.481 to 0.383. Hence the supervision is useful, but the parameterization couples two incompatible roles: one scalar must preserve endpoint termination while also suppressing the isotropic Gaussian's front tail. We reject M38 without changing its margin or weight and next isolate composition from authority.

That frozen M39 isolation resolves the apparent contradiction. We keep the M38 masses and all Gaussian geometry fixed, but interpret their weighted energy as a directly normalized categorical surface-return distribution instead of additive optical thickness. Relative to the original unit-energy reference, all/hazard/clear early returns decrease by 0.52/0.62/0.03 points, while hit recall increases by 2.17/2.32/1.46 points. All three preregistered criteria pass. An M37 descriptive arm has 0.05 points fewer all-stratum early returns but 1.57 points lower hit recall; it was not eligible for post-result selection. This experiment supports a precise representational conclusion: learned FREE/OCCUPIED/UNKNOWN mass is surface-return evidence under a categorical ray measure, not a volumetric opacity that can be summed across an arbitrarily dense primitive discretization. The result remains development-exposed pending a frozen cross-domain test.

Directly fine-tuning both authority heads through that deployed distribution does not improve the candidate. M40 raises anchor/child occupied-mass correlation to 0.494/0.616 and adds 0.69 points of hit recall over M39, but early returns rise by 1.13 points. The completion-child mean occupied mass increases from 0.383 to 0.663, while each parent has four children. Thus the ray objective exploits a family-amplitude degree of freedom: it transfers total surface measure to the denser completion family. We reject the checkpoint despite its better evidential ranking and next conserve the separately supervised anchor and child measure before allowing within-family redistribution.

M41 enforces that conservation to within $1.53 \times 10^{-5}$ per Actor, yet fine-tuning still raises early returns by 1.28 points over M39 while adding 1.07 points of hit recall. Anchor/child evidence correlations improve to 0.524/0.613. The remaining mismatch is therefore not total measure or identifiability: maximizing a nominal target-bin likelihood does not directly control whether the deployed CDF median crosses the early-return boundary. This motivates supervision on the ordered probability events used by the physical evaluation itself.

M42 consequently optimizes two GT-defined interval events under the same conserved representation: survival before $d_{\mathrm{gt}} - 0.20\,\mathrm{m}$ and return within $\pm 0.20\,\mathrm{m}$ of the observed endpoint. Aggregate early returns are 0.05 points below the unit-energy reference and hit recall is 2.54 points higher; hazardous early returns also improve by 0.11 points. However, clear-stratum early returns increase by 0.22 points, and the predicted early-event probability itself rises slightly during training while only the hit-band likelihood improves. We reject M42 under the frozen worst-stratum rule. This negative result is informative: defining physical consistency from GT events is necessary, but a global interval objective does not by itself remove stratum-dependent ambiguity in a fixed geometry and surface measure. We therefore stop development-set loss tuning and retain the simpler frozen M39 representation for one cross-domain evaluation.

M44 then applies the exact accounting in Eq. 17. Across all 99,208 rays, the CDF condition and rendered early decision agree exactly, and anchor–child decomposition has maximum residual $2.98 \times 10^{-7}$. M39 lowers mean preboundary mass by 0.331 points relative to unit energy: observed anchors contribute $-0.346$ points, whereas completion children contribute $+0.016$ points. For clear Actors, child mass instead rises by 0.163 points and cancels much of the 0.265-point anchor reduction. Thus the supported aggregate change is not a completion filter: it is driven by supervised anchor authority, while the remaining clear-stratum sensitivity is localized to child surface measure. Early and non-early rays have mean boundary CDF 0.827 and 0.114, respectively, furnishing a direct explanation of safety margin without creating a new deployment gate.

Finally, M45 replaces only the completion-child isotropic kernel by the frozen M11 GT-supervised oblate kernel. Relative to M39, all and clear early returns fall by 0.02 and 0.28 points, and hit recall rises by 3.87 points; the mean anisotropy is 8.10 with $0.020\,\mathrm{m}$ normal thickness. Hazardous early returns nevertheless rise by 0.034 points, so the frozen worst-stratum criterion rejects M45. This isolates a train–renderer mismatch rather than refuting planar support: M11 normals were learned for earliest hard intersection, while M45 uses them inside a global categorical CDF. We do not tune thickness; any successor must supervise orientation through that deployed CDF while freezing geometry and authority.

M46 does so, updating only the normal/thickness output rows with GT not-early and hit-band events. Training safe and hit NLL both decrease, and relative to M39 the final model lowers all/clear early returns by 0.05/0.59 points while adding 4.32 points of hit recall. Hazardous early returns instead increase by 0.062 points—more than the frozen M45 increase. We therefore reject and close oriented-support training. Renderer alignment is not sufficient when a local support degree of freedom must also absorb motion-, pose-, or incidence-dependent global ordering; the next analysis keeps these factors separate rather than feeding the hazard label to geometry.

M47 separates those factors without further training. The small aggregate early reduction hides 2.996 points of newly early rays and 3.044 points of repaired rays. Motion is not the explanation: hazardous moving Actors improve by 0.015 points, whereas hazardous quasi-static Actors regress by 0.809 points, largely from one trailer and one truck. Prior KEEP/PROJECT provenance does not split the failure. Grazing and oblique rays improve by 0.832/0.483

Figure 4. Frozen M24 view, cropped identically. Left to right: target image, original 32,522-primitive StreetGS Actor, and the 309-primitive geometry-locked carrier. Attribute transfer is rasterizer-compatible but loses appearance capacity; geometry is not moved to repair the image.

points, while near-normal rays regress by 0.887 points; nevertheless, actor-level correlation between incidence and the early delta is only 0.052, and all five tested physical associations have absolute correlation at most 0.100. We therefore reject motion masks, provenance filters, and incidence thresholds. M45–M47 instead expose heterogeneous ray-label transport: local support is being asked to represent canonical shape, visibility, and termination ordering simultaneously. The next representation makes visibility an explicitly supervised factor of the joint return measure; temporal deformation remains a separate trajectory-supervised problem.

**Qualitative protocol.** Main-paper and supplement scenes are frozen from metadata and registered failure strata before rendering method outputs. Panels show RGB, LiDAR, physical surface, ray evidence, compiler action, and the repaired world. We report all frozen cases, including failures, rather than selecting only visually favorable examples.

## 6. Limitations

The original AV2 attribution is a target-nearest proxy. A consumed correction and prospective disjoint AV2 read establish the operator gap, but the latter transfers a metric and provenance—not a learned repair policy. Point deletion certifies non-increasing early returns, not hit retention, Chamfer, collision freedom, or policy safety. Sparse sensing leaves unknown space; tracked cuboids may be wrong; only 64.0% of Actors add no observed-ray violation. Abstention can sacrifice geometry or hazardous-Actor coverage. The 29-Actor source fit, 8.9%→75.6% coverage shift, rank reversal, and 49.5% AV2 sensor-opportunity sensitivity rule out cross-domain exchangeability. Reliability alignment has five selected test Actors, while composition reuses source outcomes without vehicle dynamics. Static topology, deformable Actors, measured sensor-error sets, closed-loop dynamics, and appearance-dependent hazard remain open; HARP-3D is not a real-road safety guarantee.

The learned study remains development-only. M0 suppresses completion through UNKNOWN, M1 extracts no learned surface, and B4 improves surface metrics while more than doubling hazardous early returns. M7 avoids deployment filtering and passes the same development contract, but its 66-Actor fold has prior parent-representation exposure and its differentiable free-space proxy remains $1.109\times$ the M5 reference even while literal early counts improve. Finite primitive fields subsequently trade free-space precision against surface coverage. M18 avoids post-hoc filtering by supervising a one-hot GT return distribution, but its clear subset regresses and its unit-mass categorical support assumes one return inside the Actor box. It neither models ray drop nor proves a watertight surface. M19 further shows that a trainable ray field can compensate for degraded anchors, while M20 shows that a decoder-free isotropic Gaussian energy can lower its loss by inflating support scale. The resulting M21 candidate is view-independent in Actor coordinates and uses frozen GT-supervised geometry, but it was selected by a development diagnostic; its scale has no calibrated occupancy-probability meaning and its AV2 aggregate is pending. We do not use source-final or partial AV2 results to choose branch count, bins, thresholds, losses, seeds, residual bounds, or encoder adaptations; zero-shot and safety claims remain withheld until the frozen 20-log aggregate completes. The current model covers rigid annotated vehicles; deformable objects and image-conditioned geometry remain open. Exact SE(3) energy equivariance validates coordinate plumbing only: it does not establish trajectory accuracy, static/dynamic discovery, occlusion correctness, collision freedom, or improved geometry. The M29 GT tail loss improves target coverage and mean Chamfer but worsens the reverse surface tail and hazardous early returns; therefore neither mean distance nor a symmetric top-tail surrogate supplies a worst-case clearance certificate. The M30 known–possible interval preserves uncertainty without a deployment filter, but brackets only 56.69% of development returns, leaves 47.03% of upper endpoints unbounded, and exposes 19.58% anchor-only early returns. Its set-order theorem is conditional on a completion coverage assumption and cannot repair noisy canonical fusion, pose, annotation, or visibility. M31 attributes those returns but does not causally identify their error source: kept endpoints remain the majority, projected endpoints are disproportionately harmful, and aggregate moving/static rates are similar. These exposed development diagnostics cannot validate a correction or external generalization. A valid successor must alter the GT-derived training signal and retrain the field; provenance- or time-based inference deletion would not establish physical consistency. M32 further shows that replacing binary authority with soft mass is insufficient when its build input retains only canonical position, scale, primitive type, and Actor size. Although the evidential target is native-ray supervised, the input omits the source-frame and multiview evidence needed to predict that target. Its negative result does not reject evidential occupancy; it rejects the current information interface. M34 repairs that interface and obtains positive held-out occupied-mass correlation, but still fails the early-return metric. Thus evidential identifiability alone is insufficient: a spatial-energy softmax does not enforce the ordered survival semantics of a first-return sensor. A transmittance implementation remains a development hypothesis until evaluated under the same frozen ray protocol. Likewise, nearest same-Actor SH/opacity transfer preserves the carrier by construction, but a frozen native-rasterizer view loses 11.37 dB footprint PSNR when 32,522 visual primitives are replaced by 309 physical carriers. Attribute-only image supervision recovers 1.08 dB on six held-out views while leaving an 8.41 dB gap to the original Actor. This separates an available learning signal from insufficient one-carrier capacity; it does not establish photorealism, semantic fidelity, or cross-scene appearance generalization. A supervision-derived nine-surfel visual layer improves pooled PSNR by another 0.63 dB, but only one of six held-out views improves over the one-carrier model. A fixed coarse–fine hierarchy makes all six directions positive, yet those views were already exposed before hierarchy design and the original-Actor gap remains 7.62 dB. This is a representation mechanism diagnostic, not fresh view-uniform fidelity. A learned visual residual shell would still require training-time 3D support constraints before it could be claimed geometry-consistent. The scene-specific early-return direction is also negative.

# 7. Conclusion

HARP-3D separates validity, hazard, and task authority. Literal audits expose proxy undercounting and deletion monotonicity—not road-safety guarantees. A continuous-displacement oracle and several failed parameterizations show that neither UNKNOWN deletion nor unconstrained field extraction establishes physical consistency. Defining completion from motion-corrected Actor targets changes that endpoint: a small set-to-set child surface improves first return, hit recall, and Chamfer together on an exposed development fold while preserving every Actor and using the same surface during training and deployment. Its frozen AV2 run determines whether this supervision-native result transfers; until then we claim a corrected objective and source-side mechanism, not domain invariance. Physical plausibility requires jointly trained and audited geometry and ray evidence, not a single improved metric or a post-hoc filter.

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